The Coherence Infinity Generator (CIG)

Structure-first coherence — the inverse of CGI

The **Coherence Infinity Generator (CIG)** is a deterministic recursive geometric system that produces structures in which boundary relevance and internal mixing coexist across scale. The system is purely geometric and makes no assumptions about physics, biology, or cognition.

Unlike surface-first approaches (e.g., computer-generated imagery), which impose visual coherence from the outside, the CIG produces coherence as a consequence of structure itself. In this sense, it can be understood as an **inverse of CGI**: structure-first coherence rather than image-first realism.

Core Result. Across recursion depth and dimensional lift (2D $\rightarrow$ 3D), the CIG maintains non-zero boundary relevance while simultaneously increasing internal accessibility and mixing. This coexistence survives controls, depth sweeps, entropy analysis, and robustness checks.

Scope. This work does not claim to model spacetime, consciousness, or fundamental physical law. It demonstrates a structural result: recursive geometry alone can generate systems that scale through coherence rather than domination.

1. Motivation & Intuition

Many physical, biological, economic, and social systems are commonly modeled as either boundary-dominated (rigid, protected, constrained) or bulk-dominated (diffuse, mixed, unconstrained). Such dichotomies struggle to explain systems that must scale while preserving both integrity and adaptability.

The CIG was developed to explore whether recursive structure alone can naturally generate systems that support both properties simultaneously, without tuning or external control.

This question is structural rather than domain-specific: if answered positively, it would suggest that coherence across scale may arise from geometry itself.

2. The Coherence Infinity Generator

The generator begins with an infinity-like base curve and applies deterministic recursive rules: pocket placement within lobes, shell-based distributions, and increasing recursion depth. In three dimensions, shells become spherical layers and recursion produces a filamentary volumetric geometry rather than a solid fill.

> **Importantly, no parameter is tuned** to enforce boundary dominance or bulk mixing. Both emerge from the recursive construction itself.

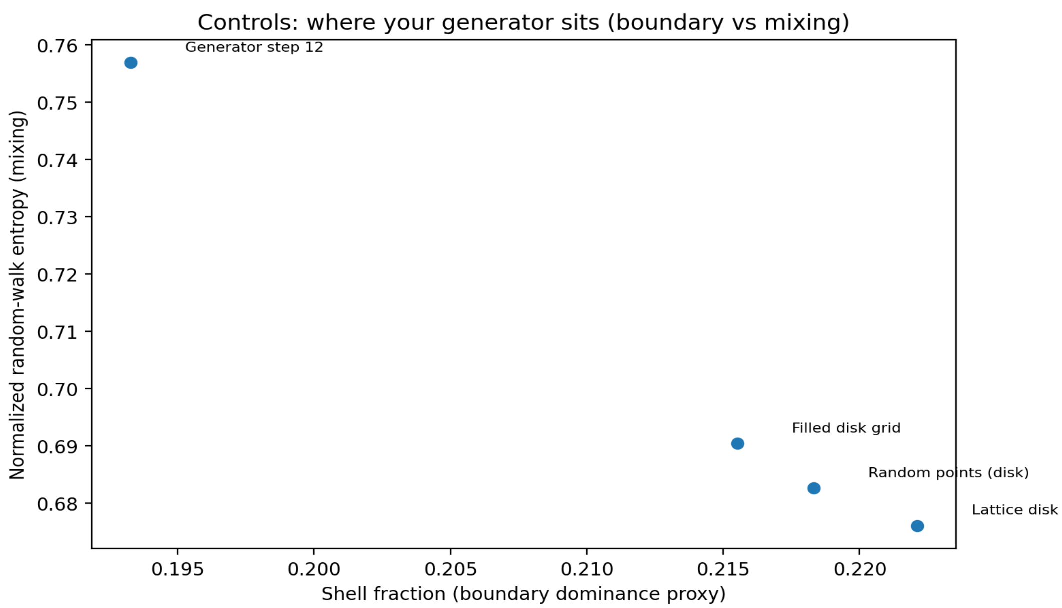
3. Measurement Framework

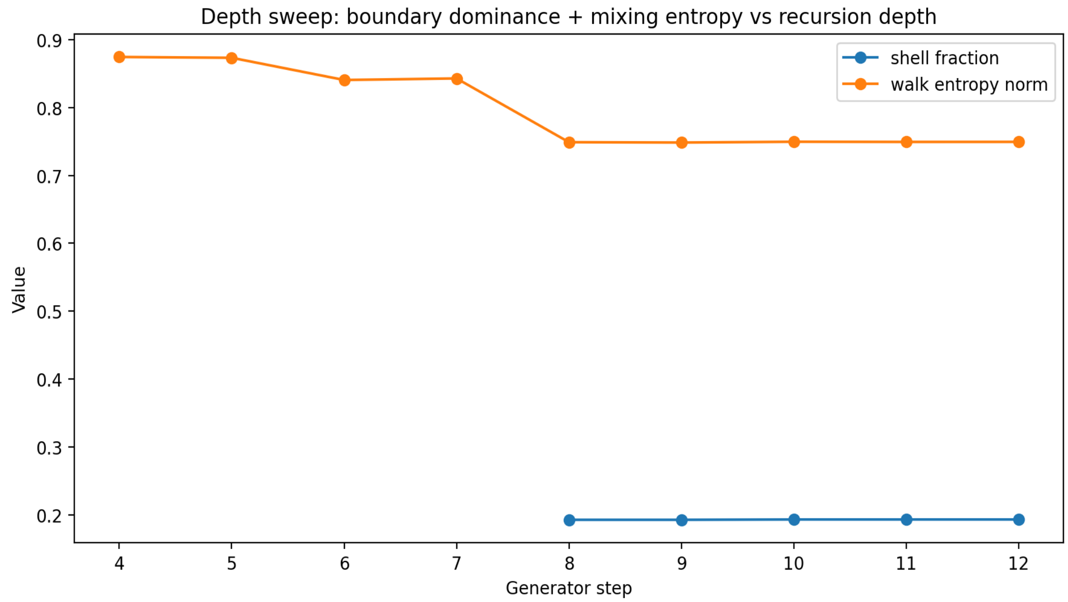
Structure is quantified using two complementary measures: (i) boundary vs bulk decomposition and (ii) graph-based random-walk entropy as a proxy for internal accessibility.

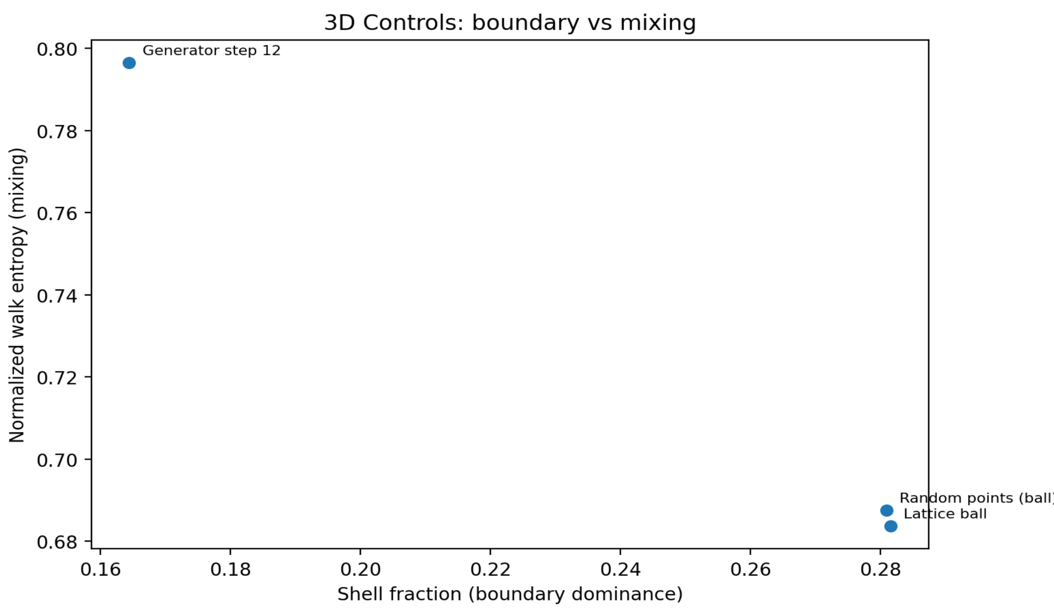
For a region of radius R, complexity is approximated as $\mathbf{K(R) \approx aR^2 + bR^3 + c}$ in three dimensions (or $aR + bR^2 + c$ in two dimensions), where coefficients capture surface-like and volume-like contributions respectively.

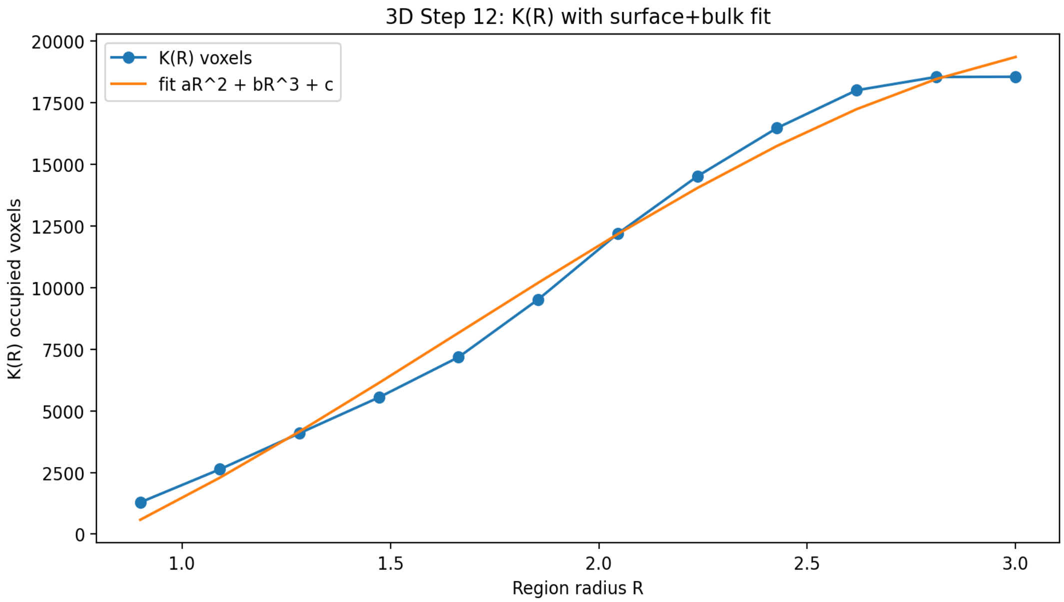
Internal mixing is measured by initiating random walks on adjacency graphs constructed from discretized representations of the structure and computing normalized entropy over walk endpoints.

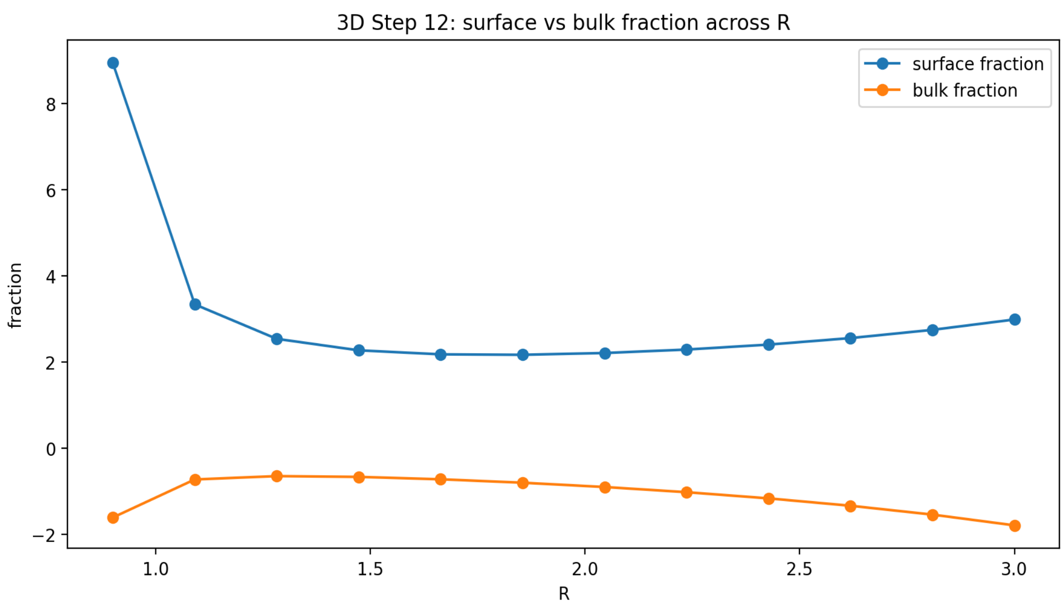
4. Results Overview (2D → 3D)

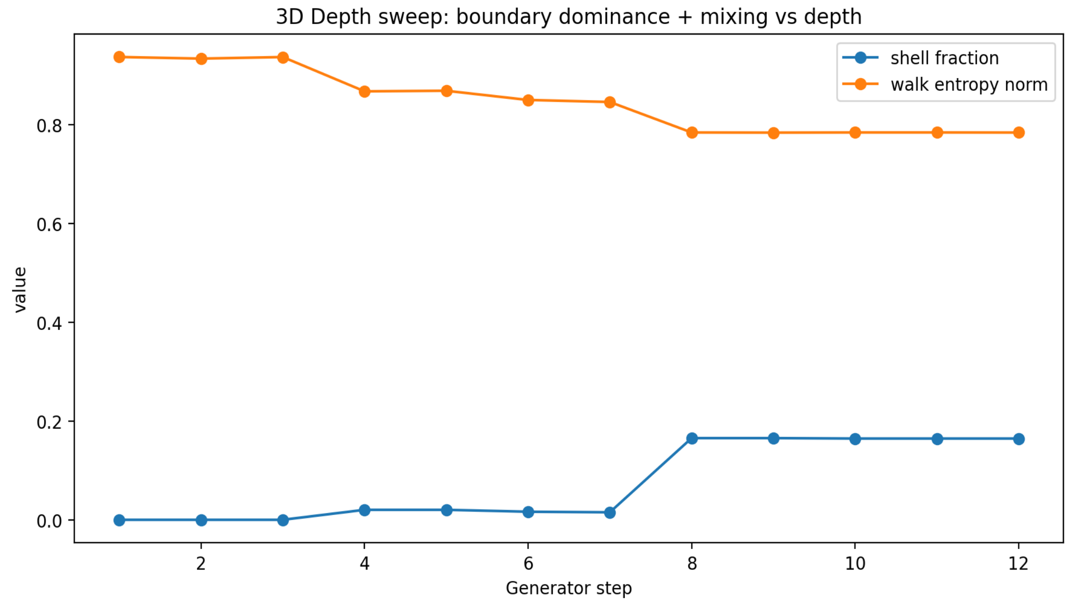


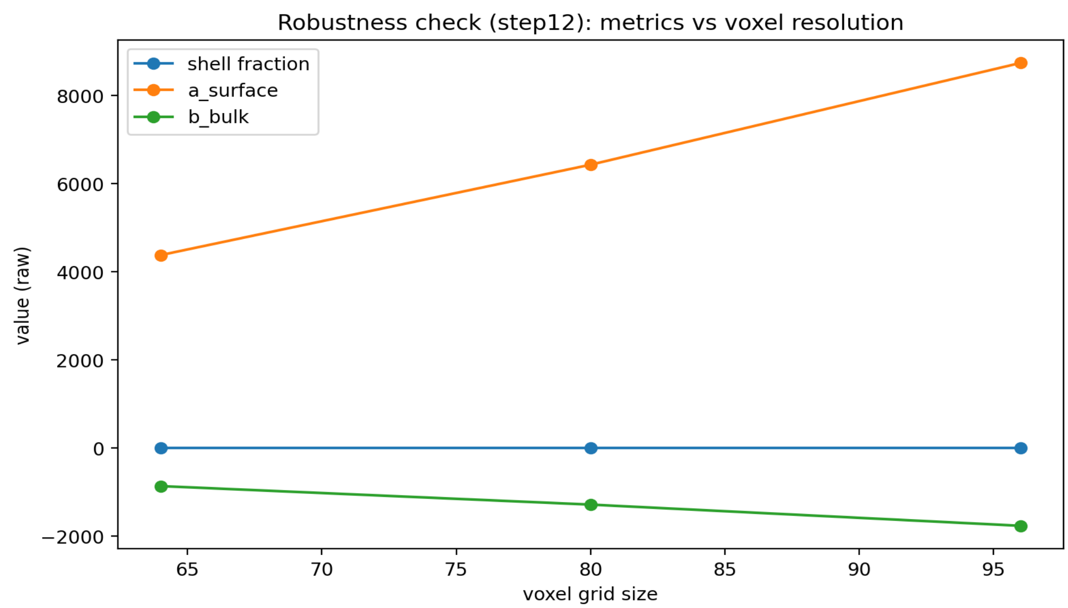

Depth sweep: boundary dominance + mixing entropy vs recursion depth
shell fraction
walk entropy norm
Value
Generator step
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
4
5
6
7
8
9
10
11
12


3D Controls: boundary vs mixing
Generator step 12
Random points (ball)
Lattice ball
Normalized walk entropy (mixing)
Shell fraction (boundary dominance)
0.80
0.78
0.76
0.74
0.72
0.70
0.68
0.16
0.18
0.20
0.22
0.24
0.26
0.28


3D Step 12: K(R) with surface+bulk fit
K(R) voxels
fit aR^2 + bR^3 + c
K(R) occupied voxels
Region radius R
0
2500
5000
7500
10000
12500
15000
17500
20000
1.0
1.5
2.0
2.5
3.0


3D Step 12: surface vs bulk fraction across R
surface fraction
bulk fraction
fraction
R
−2
0
2
4
6
8
1.0
1.5
2.0
2.5
3.0





5. Interpretation & Limitations

Across both two and three dimensions, the CIG occupies an intermediate regime between boundary-dominated and bulk-dominated controls. Boundary relevance remains non-zero while internal mixing increases with recursion depth.

These results do not establish identity with physical spacetime or biological systems. They provide a concrete example of a structure-first mechanism capable of generating properties often attributed to fundamental laws.

Limitations include discretization effects, proxy-based entropy measures, and finite recursion depth. Nevertheless, the qualitative signature persists across controls, dimensional lift, and robustness checks.

Appendix: Applied Instance — Economic Coherence (BREZ)

An analogous structure appears in the BREZ Infinite Growth Generator, where recursive capital loops must remain coherent across demand creation, realization lag, and reinvestment. In both systems, failure is phase-based rather than gradual, arising when lagged boundaries are exceeded.

Coherence Infinity Generator (CIG)		**BREZ Infinite Growth Generator**
Recursive Geometry	→	Recursive Capital Loops
Pocket Recursion	→	Loop A: DTC Demand
Ring / Shell Distribution	→	Loop B: Retail (Lag)
Stacked Layers	→	Loop C: Retention / Reinvest
Boundary + Bulk Coexist	→	Working Capital Boundary
Entropy = Accessibility	→	Cash Velocity / Optionality
Phase Threshold (Click)	→	Sudden Crash (Lag Trigger)

Same Structural Law:
Reinforcing Loops + Lag + Boundary → Compounding or Phase Flip

This correspondence suggests that stable compounding is governed by structural coherence rather than domain-specific rules, enabling cross-domain reasoning about growth and resilience.